

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (EC) Semester - IV Examination, 2011
EC – 208 Analog Electronic Circuits

Date: 12/05/2011, Thursday

Time: 1:00 p.m to 4:00 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in separate answer books.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION –I

Q-1 Answer following Questions [14]

- (a) What is linear time base generator? Explain Transistor constant current sweep. 8
- (b) Define current controlled negative resistance devices. Explain Mono-stable operation of negative resistance devices. 6

OR

Q-1

- (a) Draw and explain formation of sweep in transistor bootstrap time base generator. 8
- (b) Define voltage controlled negative resistance devices. Using diagram explain bi-stable operation of negative resistance devices. 6

Q-2 Attempt following Questions [6]

- (a) In brief, explain the linearity correction technique in current time base generator 3
- (b) Draw neat diagram of R-2R ladder type Digital to Analog Converter 3

OR

Q-2

- (a) Using diagram illustrate concept of miller time base generator for linearity improvement for time base generator 3
- (b) Draw neat diagram of weighted resistor type Digital to Analog Converter 3

- Q-3 Attempt Any Three [15]**
- (a) Tracking Type Analog to Digital Converter
 - (b) Tunnel Diode Characteristics
 - (c) Simple Current Sweep Generation
 - (d) Class A Amplifier Operation

SECTION -II

- Q-4 Answer all questions [08]**

- (a) Mention advantages of negative feedback amplifier
- (b) Explain Barkhausen Criteria for oscillations
- (c) Explain working principle of Crystal Oscillator
- (d) In brief, explain CMRR terminology

- Q-5 Answer following Questions [09]**

- (a) Explain Transistor Clipper 5
- (b) Draw the feedback amplifier topologies for Voltage Series, Voltage Shunt, Current Series and Current Shunt Feedback Amplifiers 4

OR

- Q-5**
- (a) Explain IC Voltage Regulators 5
 - (b) Derive the equations for input resistance (R_i) for voltage series, voltage shunt, current series and current shunt feedback amplifier 4

- Q-6 Attempt any three [18]**

- (a) State and prove Clamping Circuit Theorem. In addition, with help of neat sketches explain negative clamper circuit.
- (b) Explain Wien Bridge Oscillator
- (c) Explain biased clipping at two independent levels
- (d) Perform circuit analysis of BJT differential Amplifier